

Sound Patterns of Syntax: Object Shift*

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Abstract

In this paper, I explore what a purely phonological account of object shift (OS) involves and what research questions it leads to, in particular what it means for word order to be phonologically motivated and what morpho-phonological primitives are involved. I pursue the possibility that what licenses OS of full DPs in Icelandic is phonological properties, not found in other Scandinavian languages, together with overt case marking. Although the position of the object is determined phonologically, the architecture I propose, in which topic and focus play a central role, allows for an account of the semantic effect associated with OS in Icelandic.

1. Introduction

The phenomenon of Object Shift (OS) in general and Holmberg's Generalization in particular have been intriguing to linguists working within the Minimalist Program since Holmberg, 1986 in view of the restriction (in Scandinavian languages) of OS to structures that have undergone verb-movement. This type of restriction is problematic since there is no obvious way of linking the occurrence of one rule to the occurrence of another. Another puzzle raised by this phenomenon is its differing prop-

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1 erties across Scandinavian languages: In Mainland Scandinavian (MSc)
 2 and Faroese, only weak pronouns shift, yet in Icelandic, full DPs shift as
 3 well. Finally, OS has interpretive effects: only topics or non-foci shift.
 4 Holmberg, 1999 argues that OS applies in a post-spell-out component of
 5 stylistic rules, containing rules dependent on focus structure. Movement
 6 is triggered by the need of –foc constituents to be governed by a +foc
 7 feature and is only licensed when all phonologically visible non-adjunct
 8 material has been removed from between the launching site and the land-
 9 ing site of the movement. Holmberg derives the linkage between verb
 10 movement and OS from the assignment of an inherent +foc feature to
 11 verbs. The restriction on OS in terms of phonological visibility calls for
 12 a phonological account, yet Holmberg rejects the idea that OS is a pho-
 13 nological or a morphological operation since it does not make reference
 14 to phonological or morphological primitives. Chomsky, 2001b, although
 15 tempted by a phonological analysis of OS, rejects it at least in part pre-
 16 cisely because it is driven by the semantic interpretation of the shifted ob-
 17 ject. Bobaljik, 2002 proposes a copy theory of movement which allows
 18 for either copy to be pronounced. OS occurs obligatorily in the syntax
 19 yet morphological adjacency constraints determine which copy is pro-
 20 nounced at PF. This allows for a syntactic account of OS which is sensi-
 21 tive to PF properties such as adjacency.

22 In this paper, I explore what a purely phonological account of OS
 23 involves and what research questions it leads to, in particular what it
 24 means for word order to be phonologically motivated and what morpho-
 25 phonological primitives are involved. I pursue the possibility that it is the
 26 special properties of Icelandic word and compound stress, not found in
 27 the other Scandinavian languages, together with overt case marking,
 28 which licenses OS of full DPs. I also appeal to the grammatical model
 29 proposed in Erteschik-Shir, 1997, which allows for the association of
 30 phonological processes with semantic effects. I start by examining the
 31 free ride approach to OS in which the object prosodically incorporates
 32 into the verb and moves along with it (section 2.1) and in so doing un-
 33 cover some of the restrictions which phonological displacement must be
 34 subject to. Although Holmberg's generalization falls out automatically
 35 from this approach, it must nevertheless be rejected because it leaves cases
 36 of OS in which the object is not attached to the verb unaccounted for.
 37 Section 2.2 introduces a processing constraint which requires that

1 argument roles be overtly identified and shows how adverb placement is
 2 one way of overtly identifying the subject. Since processing constrains the
 3 position in which the adverb can occur, positioning of the adverb must
 4 be phonological if we take PF to be concerned with both production and
 5 processing. Yet section 2.2 shows that adverb displacement does not ad-
 6 here to the phonological principles set out in section 2.1. Section 2.3,
 7 inspired by Áfarli 1997, proposes a mechanism of adverb linearization
 8 which is compatible with these phonological principles and takes into
 9 account that not only must object pronouns prosodically incorporate,
 10 but so must these adverbs. Since prosodic incorporation plays an impor-
 11 tant role in the account of OS, section 2.4 explores this phonological pro-
 12 cess more carefully. Section 3 outlines how a phonological approach has
 13 the potential to explain language variation, including fine distinctions
 14 between dialects. Section 4 argues that it is Icelandic phonology which
 15 accounts for full DP OS in this language. Section 5 provides a brief com-
 16 parison with Bobaljik, 2002; his proposal is designed to allow for phono-
 17 logical restrictions and in this way comes closest to my own. The last
 18 section outlines an architecture which can accommodate a phonological
 19 account of word order.

20

21

22 2. The phonological basis of Object Shift

23

24 2.1. First Pass

25

26 The linkage of OS to verb-movement is illustrated in (1) and (2) (all
 27 examples are in Danish unless otherwise specified):

28

(1) Peter mødte ham ikke.

29

Peter met him not

30

‘Peter didn’t meet him.’

31

(2) a. Peter har ikke mødt ham.

32

Peter has not met him

33

b. ... at Peter ikke mødte ham.

34

that Peter not met him

35

36 In (1) both the verb and the object have moved. In the sentences in (2),
 37 neither the verb nor the object have moved. In (2)a, the presence of the

1 auxiliary blocks verb-movement and (2)b illustrates the lack of verb
2 movement in subordinate clauses.

3 In Erteschik-Shir, 2001; Erteschik-Shir and Strahov, 2004, it was
4 argued that the object piggy-backs on the verb and therefore moves only
5 when the verb moves. OS as such does therefore not occur.¹ This
6 approach (also known as the ‘free ride’ approach; see Hellan, 1994),
7 automatically captures Holmberg’s generalization since if the object at-
8 taches to the verb, it will only move when the verb moves. In order to
9 see how the free-ride approach does in fact offer a solution to examples
10 such as this one, it is necessary to examine the implementation of this
11 approach. An obvious tack to take is to view the weak pronouns as
12 undergoing cliticization (as in Hellan and many others especially Josefs-
13 son, 1992; 1993, and Deprez, 1994). Such an account is however not ex-
14 tendable to Icelandic, in which full DPs shift and so cannot serve as an
15 explanation for OS as a whole. In the papers cited above, I therefore
16 proposed a different form of attachment, prosodic incorporation (PI), a
17 phonological process which forms a single prosodic unit out of the pro-
18 sodically incorporated element and its host. What was proposed there
19 was an extension of cliticization processes not only to weak pronouns
20 but also to other constituent types. This more general incorporation pro-
21 cess is what I refer to as PI. I outline the phonology of PI in section 2.4.

22 If prosodic incorporation is a phonological process, and if, as I pro-
23 pose, it feeds verb movement, then verb movement must also be a phono-
24 logical process. (This is consistent with Chomsky, 2001b, who views head
25 movement as post-syntactic.) Assume that strings such as (3) are, very
26 roughly, the output of syntax.²

27 (3) [ikke Peter mødte ham]
28 not Peter met him
29

30 I do not take a position here on whether the syntax feeds phonology
31 in chunks or phases (Chomsky, 1999) and whether these chunks are
32

33
34 ¹ Under the proposed account, the object does not shift. For convenience I still refer to
35 the phenomenon as OS.

36 ² I assume late merger of inflectional morphology, but nothing here depends on exactly
37 how it is implemented. For convenience, I list the inflected forms in the phonological
string.

1 syntactically or phonologically motivated. I limit discussion to the larger
 2 VP domain within which all arguments are merged. Together with most
 3 researchers, I also assume (for the moment) that negation and other VP
 4 adverbials are merged at the (left) boundary of the VP.³ For the time be-
 5 ing, I take it that only the linear string and its boundaries are visible to
 6 phonological processes.

7 The phonological derivation of (1) is shown in (4).

- 8 (4) a. [ikke Peter mødte ham]
 9 b. PI: [ikke Peter mødte+ham]
 10 c. V-2: [mødte+ham ikke Peter]
 11 d. top: [Peter_{top} mødte+ham ikke]

12 (4)a is the output of PI, where PI is indicated by '+'. PI applies to weak
 13 functional elements such as pronouns and auxiliaries and is constrained
 14 by language particular prosodic rules.⁴ In the system envisaged here, PI
 15 is optional, yet non-PI-ed weak elements are ruled out as being un-
 16 pronounceable, i.e., uninterpretable by the PF interface. Once PI has
 17 occurred, the host and the PI-ed element together form a phonological
 18 constituent and will therefore move as a unit as shown in (4)c. Here the
 19 movement of the verb (or auxiliary) to second position (V-2) is im-
 20 plemented as phonological displacement to the left edge (indicated by
 21 the bracket).⁵ Topicalization applies to the subject, moving it to initial
 22 position resulting in (4)d.⁶ Disregarding details the derivation is disarm-
 23 ingly simple. And the examples in (2) also follow since there the pronouns
 24 incorporate in situ.

25 A potential problem for the free ride analysis of OS is posed by
 26 the seeming need for rule ordering. This would make the approach ex-
 27 tremely unattractive. Rule ordering, however, need not be imposed. The
 28

29
 30 ³ Section 2.3 presents a different account of adverb positions.

31 ⁴ See Selkirk, 1996 for an analysis of the incorporation of function words.

32 ⁵ For the moment I disregard the possible motivation for phonological dislocations and
 33 hypothesize that they are triggered by pronunciation and processing needs. I return to
 34 this issue in section 6.2.

35 ⁶ See Erteschik-Shir, 2003, submitted for an architecture in which top and foc features are
 36 introduced lexically.

37 I discuss the motivation for fronting the subject in section 2.3 and conclude that the
 subject in fact stays in situ. The word order of is achieved, I propose, by implementing
 constraints on adverb pronunciation.

1 derivations in (5) show that other potential orders result in unpronounce-
2 able strings.

- 3 (5) a. [ikke Peter mødte ham]
4 b. V-2: [mødte ikke Peter ham]
5 c. top: [Peter mødte ikke ham]
6 d. PI: does not apply
7 (6) a. [ikke Peter mødte ham]
8 b. V-2: [mødte ikke Peter ham]
9 c. PI: [mødte ikke Peter+ham]
10 d. top: [Peter+ham mødte ikke]
11 (7) a. [ikke Peter mødte ham]
12 b. PI: [ikke Peter mødte+ham]
13 c. top: [Peter ikke mødte+ham]
14 d. V-2: does not apply
15

16 In (5), V-2 precedes PI leaving the pronoun unpronounceable: weak
17 pronouns cannot be pronounced on their own and in Danish, adverbials
18 do not provide a host for incorporation. (Note that the array of hosts is
19 language specific and may vary even across dialects. Also, as we shall see
20 below in section 3, some dialects of Swedish, and even Danish, do allow
21 the pronunciation of weak pronouns following negation.)

22 In (6), PI precedes topicalization and the object incorporates into the
23 subject, a possible host as we shall see below. Topicalization of the sub-
24 ject will drag along the incorporated object resulting in (6)d. This string
25 violates a processing constraint which blocks elements with ϕ -features
26 from intervening between the subject and its verb.⁷

27 (7) illustrates the case in which topicalization precedes V-2. Here V-2 is
28 blocked from applying since the target of a P-syntactic process must be a
29 string edge and therefore the verb could only move to first position, and
30 not to second position as required.

31 So far, the proposed phonological implementation can be made to
32 work without imposing rule ordering. We have also proposed the follow-
33 ing ground rules for phonological displacement (D):
34
35

36 ⁷ This constraint is subsumed under a more general processing constraint on identification
37 of arguments (ID) introduced in (16).

1 (8) Properties of D

- 2 a. D is to an edge.
3 b. Ds are unordered.

4 I also adopt the hypothesis that motivation for D is limited to easing
5 pronunciation (production) and parsing (processing) as stated in (9):⁸

6 (9) Motivation for D

- 7 a. production
8 b. processing

9
10 Restrictions on pronunciation will vary vastly across languages and de-
11 pend on the general phonological properties of the language. These, as
12 will be shown below include, stress and prosodic and intonational phras-
13 ing. Restrictions on processing will depend on disambiguating factors.
14 Among the disambiguating factors to be discussed are overt case mark-
15 ing, overt agreement and the fronting of topics. Since the motivation
16 for phonological displacement is ease of pronunciation and parsing, and
17 neither is an absolute value, at least in some cases displacement will be
18 optional.

19
20
21 2.2. *Second Pass*

22
23 An immediate problem with this approach (and indeed the reason it
24 was rejected by Hellan), is the fact that the free ride approach leaves
25 unaccounted for cases of OS in which the object is not attached to the
26 verb.⁹ (10) illustrates such a case.

- 27 (10) I går mødte Peter ham ikke
28 Yesterday met Peter him not
29 'Yesterday Peter didn't meet him.'

30
31 In (10) the object has shifted, but clearly not with the verb since the sub-
32 ject intervenes between the verb and the shifted object. (11) shows that the
33 derivation of such sentences requires additional assumptions.

34
35 ⁸ I ignore morpho-phonological constraints requiring adjacency between roots and affixes.
36 In the late insertion model I envisage, affixation does not require movement.

37 ⁹ For this point see also Bobaljik, 2002, p. 26.

- 1 (11) a. [ikke Peter mødte ham i går]
 2 b. V-2: [mødte ikke Peter ham i går]
 3 c. PI: [mødte ikke Peter+ham i går]
 4 d. top: [i går mødte ikke Peter+ham]
 5 e. subj?: [i går mødte Peter+ham ikke]

6 Up to and including (11)d the derivation is very much as before. Note
 7 that if PI precedes V-2 (as in (4)), the output will be (12) in which an ele-
 8 ment with φ -features intervenes between the subject and its verb.

- 9 (12) [i går mødte+ham Peter ikke]
 10

11 This constraint holds whether the subject is to the left of the verb as in
 12 (6)d or whether it is to its right as in (12). Yet the movement of the sub-
 13 ject in (11)e poses a problem since it looks like the subject (with its in-
 14 corporated object) moves to a non-edge position. Such movement is
 15 excluded in the phonology according to the ground rules in (8) and it is
 16 not clear what motivation there would be for such a move even if it were
 17 not excluded. An obvious alternative is moving the adverb to the right.
 18 Phonological displacement is not directionally restricted and movement
 19 of the adverb to the right would be to an edge position. Yet what would
 20 be the motivation for moving the adverb? Since the unmoved adverb is
 21 positioned between the verb and its subject, the motivation could be to
 22 enable adjacency between the two, yet such adjacency is not required in
 23 subordinate clauses such as (13).

- 24 (13) ... at Peter ikke mødte ham.
 25 ... that Peter not met him.
 26 '... that Peter didn't meet him.'

27 Here negation intervenes between the subject and the verb to no ill effect.
 28 It follows that there is no adjacency requirement between subject and
 29 verb and that adverbs (but not elements with φ -features) can intervene.
 30 There are two types of motivation for phonological movement, either
 31 parsing ease or pronounceability. Let us examine each one in turn.

32 According to Raviv, in prep. adverbials, auxiliaries and modals make a
 33 difference in the potential for topicalization of full DPs in Danish:

- 34 (14) a. *Peter mødte Sara
 35 Peter met Sara
 36 'Sara met Peter.'
 37

- 1 b. Peter mødte Sara ikke.
 2 Peter met Sara not
 3 'Sara didn't meet Peter.'
 4 c. Peter mødte ikke Sara.
 5 Peter met not Sara'
 6 'Peter didn't meet Sara.'
 7 (15) a. *Peter kender Sara fra fjernsynet
 8 Peter knows Sara from TV-the
 9 'Sara knows Peter from TV.'
 10 b. Peter burde Sara kende fra fjernsynet.
 11 Peter ought to Sara know from TV-the
 12 'Sara ought to know Peter from TV.'

13 Raviv argues that the presence of the adverbial and the modal in the b.
 14 sentences prevents ambiguity as follows: The subject is identified (as a
 15 subject) due to its position preceding the adverb whereas arguments fol-
 16 lowing the adverb are identified as objects. The adverb therefore func-
 17 tions to delineate the boundary between subjects and other arguments.¹⁰
 18 Similarly the word order in (15)b is disambiguated by the presence of
 19 the modal (an auxiliary would have the same effect). If the first argument,
 20 *Peter*, were a subject, then the modal would be followed by the verb.
 21 Since it is in fact followed by another argument, *Sara*, this latter argu-
 22 ment must be the subject.

23 Raviv bases her analysis on Erteschik-Shir, 2003, submitted which
 24 introduces the following parsing constraint on the identification of argu-
 25 ments. This identification constraint applies at the PF interface with the
 26 articulatory-perceptual system:

- 27 (16) *ID*:
 28 In a string, X . . . Y, where X and Y are arguments, ID X as subject
 29 and Y as object if neither is marked otherwise.
 30

31 Thus in a string consisting of $\text{arg}_1\text{-V-arg}_2$, arg_1 would be identified as a
 32 subject and arg_2 as an object, unless the case marking on arg_1 is that of
 33

34 ¹⁰ This is of course not the case with OS in which the object pronoun precedes the adverb.
 35 Object pronouns are identified by their overt case marking and therefore need not be
 36 identified by their linear order. The same holds for full DPs in Icelandic as demonstrated
 37 in section 4.

1 an object or, alternatively, the case marking on arg_2 is that of a subject
 2 (or both). In these cases, arg_1 would be parsed as a topicalized object.
 3 This explains why the sentences in (17) can be understood with a topical-
 4 ized object whereas (14)a cannot.

- 5 (17) a. Ham mødte Sara.
 6 him met Sara
 7 ‘Sara met him.’
 8 b. Peter mødte hun.
 9 Peter met she
 10 ‘She met Peter.’

11 The verb also functions as an identifier. It identifies one of the arguments
 12 adjacent to it, as the subject. Since the subject and object can both be ad-
 13 jacent to the verb, the verb only identifies the subject unambiguously if
 14 the subject and the verb agree overtly. The ID of the subject is therefore
 15 inhibited if an object intervenes between it and the verb, explaining the
 16 restriction on intervening elements with φ -features.

17 ID would identify the topicalized object in (14)a and (15)a as a subject
 18 and the post verbal argument as an object, indeed the only interpretation
 19 of the string. In (14)b and (15)b the added adverbial and modal force a
 20 different identification thus licensing topicalization in these cases. It seems
 21 then, that in spite of the fact that the adverb seemingly should play no
 22 role in the assignment of syntactic roles to the arguments, it in fact plays
 23 a disambiguating role. Is this, then, the motivation for the movement of
 24 the adverb to the right? The answer is negative since both adverb orders
 25 render acceptable sentences, as seen in (14)b and (14)c.

26 We now turn to an examination of the phonology of Scandinavian
 27 adverbials, in order to see if the motivation for the placement of adverbs
 28 is to be found in this domain. Although some adverbials also have a free-
 29 standing (strong) form as in (18), many have only a weak form which
 30 must prosodically incorporate as in (19).¹¹

- 31 (18) a. Sandsynligvis mødte Peter ham.
 32 probably met Peter him
 33 ‘Peter probably met him.’
 34

35
 36 ¹¹ It is likely that the distinction here is between sentence adverbials which have both a
 37 strong form and a weak form and VP adverbials which have only a weak form.

- 1 b. Peter mødte ham, sandsynligvis.
 2 Peter met him, probably.
 3 c. Peter mødte+ham+sandsynligvis.
 4 (19) a. *Ikke/*Ofte mødte Peter ham
 5 not/often met Peter him
 6 b. *Peter mødte ham, ikke/ofte.
 7 Peter met him, not/often
 8 c. Peter mødte+ham+ikke/ofte.

9 The strong types of adverbs occur sentence initially and sentence finally
 10 with comma intonation as shown in (18)a and (18)b. These positions are
 11 not available for the weak adverbs as illustrated in (19)a and (19)b. (18)c
 12 and (19)c show that both forms incorporate elsewhere, i.e., in the cases
 13 under examination.

14 Although the requirement that adverbs incorporate in (18)c and (19)c
 15 does constrain the possible locations of adverbs, it does not lead us to a
 16 solution to the problem of motivating the shift of the adverb to the right.
 17 Compare the final two strings in (11), repeated in (20), with (21).
 18

- 19 (20) a. top: [i går mødte ikke Peter+ham]
 20 b. subj?: [i går mødte Peter+ham ikke]
 21 (21) Peter mødte+ikke Sara.
 22 PETER met not Sara

23 (21) shows that the adverb can incorporate into the verb. It follows
 24 that neither the adverb nor the pronoun are stranded in (20)a which
 25 therefore cannot be ruled out for reasons of pronounceability. Motivation
 26 for the movement of the adverb to the right of the subject is therefore still
 27 lacking.
 28

30 2.3. *Third Pass*

31 Although the position of the subject to the left of the adverb in (20) is
 32 motivated in terms of processing, there is no obvious way of formulating
 33 it as motivation for the rightward movement of the adverb. The problem
 34 disappears, however, if we abandon the common assumption that ad-
 35 verbs are merged at the left VP edge. Instead, following Åfarli, 1997 and
 36
 37

1 Chomsky, 2001a, Bobaljik, 2002¹², adjuncts are assumed to be merged on
 2 a separate plane. Åfarli's approach is designed to explain the possible po-
 3 sitions of adverbials, among them the order of weak pronouns and adver-
 4 bials. He analyzes weak pronouns as clitics which "must form a prosodic
 5 unit together with some other phonetically realized host adjacent (typi-
 6 cally left-adjacent) to it, where that host may be an element belonging to
 7 any category (verb, noun, adjective, preposition, complementizer, etc.) as
 8 long as it is not an adverbial" (p. 102). Although the intuition behind
 9 Åfarli's approach is similar to the one presented here, it does not allow
 10 for the prosodic incorporation of nonclitics and so does not generalize
 11 this account to OS of full DPs. This is because Åfarli's implementation
 12 is syntactic. I present it here, and examine the properties of a phonologi-
 13 cal implementation.¹³

14 According to Åfarli, adverbials are attached in a third dimension to
 15 AgrP (either Agr_sP or Agr_oP). They are then linearized in one of three
 16 ways: Either the adverbial is "bent" down, to the left, or to the right, de-
 17 riving a structure in which the adverbial ends up adjoined to Agr', or to
 18 the left or right of AgrP, respectively. In the latter case, the adverbial is
 19 sentence final and is pronounced with comma intonation. Since multiple
 20 adverbials can attach and each can bend independently, many adverb
 21 orders can be derived. Excluded, however, is the following in Swedish
 22 (from Åfarli, p. 96, his (9b)):

- 23 (22) *Har nogon student last möjligen boken?
 24 has any student read possibly the book

25 Bending the adverb down positions it after the subject; bending to the left
 26 positions it to the left of the subject, and bending to the right positions it
 27 at the end of the sentence (with comma intonation). In (22) the adverb is
 28 positioned inside the VP, not a bending option in Åfarli's system.

29 Bending, according to Åfarli, must take place before Spell-Out since it
 30 is necessary for both PF and LF interpretation. In the present approach,
 31 interpretation follows from the adjunction site of the adverbials together
 32

33
 34 ¹² The idea originates with Lebeaux, 1988; 1992 who argues for the insertion of adjuncts at
 35 s-structure. The need for deriving adjuncts in this way follows from their invisibility with
 36 respect to adjacency requirements, among them those pertaining to OS.

37 ¹³ For an initial account of linearization from a three dimensional structure to a pro-
 nounceable string see Erteschik-Shir, 1987.

1 with the f-structure of the sentence (see section 6.1) and not from the
2 linear position of the adverb. Let us therefore assume, as the null hy-
3 pothesis, that adjuncts are phonologically linearized anywhere within the
4 domain of the constituent to which they are attached, where the domain
5 of a constituent includes the positions immediately to its left, immediately
6 to its right and anywhere in between. Actual positions of adjuncts are
7 constrained by purely phonological means (allowing for a measure of
8 cross linguistic variation). In the case at hand, the VP adjunct can occur
9 VP initially, VP finally and between the VP elements. Once topicalization
10 and V-2 have occurred, we are left with the following three possibilities:

- 11 (23) a. *i går mødte ikke Peter ham
12 yesterday met not Peter him
13 b. *i går mødte Peter ikke ham
14 c. i går mødte Peter+ham+ikke

15 (23)a is ruled out because it violates the ID processing constraint which
16 requires that the subject occur to the left of the adverb leaving the subject
17 unidentified. (23)b is ruled out because the adverb is not a possible host
18 for PI and the pronoun is therefore left stranded.¹⁴ This leaves (23)c as
19 the only pronounceable string, with both the pronoun and the adverb in-
20 corporated as required. The three dimensional merger of adjuncts allows
21 for the free linearization of adjuncts within the domain of the constituent
22 to which they are attached modulo independent phonological processing
23 and pronunciation constraints. This solves the problem of motivating
24 adjunct movement since no motivation is needed if adjuncts are not as-
25 sumed to move.¹⁵

26 Adopting this account of adjunct placement also leads one to question
27 the account of the classic case of OS in (1), shown in (4). There it was as-
28 sumed that the incorporated object gets a free ride with the verb and that
29 the subject topicalizes. Yet, according to Raviv, in prep, subject topics are
30 not pronounced with the initial stress characteristic of other fronted
31 topics. The topicalized object in (24)b is stressed, but the subject topic in
32 (24)a is not.

34
35 ¹⁴ See section 2.4 for the phonological details of this account.

36 ¹⁵ See Costa, 2004 for evidence from European Portuguese that the weight of adverbs
37 determine their position in the string.

- 1 (24) a. Peter mødte+ham ikke.
 2 Peter met him not
 3 'Peter didn't meet him.'
 4 b. HAM mødte Peter ikke.
 5 him met Peter not
 6 'Peter didn't meet him.'

7 This intonational difference suggests that subject topics are in fact not dis-
 8 located in the phonology. Yet the verb does occur in second position
 9 (with its incorporated pronoun), and if phonological displacement targets
 10 edges, as I claimed above, then displacing the verb without displacing the
 11 subject by topicalization will not get us the desired results. If, however,
 12 adjunct placement can be sensitized to V-2 so that the adverbs will be pre-
 13 vented from linearizing in second position in main clauses, neither V-2
 14 not topicalization need occur in this case. Other potential positions for
 15 the adverb are the initial position, where it has no host, and between the
 16 verb and the pronoun, which leaves the pronoun stranded. The adverb
 17 must therefore linearize at the end of the clause where it incorporates.

18 If, however, adjunct placement can be sensitized to V-2, so that the
 19 adverbs will be prevented from linearizing in second position in main
 20 clauses, neither V-2 not topicalization need occur in this case. The adverb
 21 can only occur at the end of the clause since in initial position it has no
 22 host and if it intervenes between the verb and the pronoun, the pronoun
 23 is stranded.

24 The free linearization account of adjunct position however complicates
 25 the account of PI in situ invoked for the auxiliary and subordinate clause
 26 cases in (2) repeated in (25).
 27

- 28 (25) a. Peter har ikke mødt ham.
 29 Peter has not met him
 30 b. ... at Peter ikke mødte ham./* mødte ham ikke
 31 that Peter not met him
 32

33 Our account so far predicts that the adverbs can occur in the positions
 34 they do. In (25)a the adverb occurs after the auxiliary (in second position)
 35 with the subject to its left. It is pronounced incorporated into the auxil-
 36 iary and so no pronunciation or processing constraints are violated. Sim-
 37 ilarly, in the subordinate clause in (25)b the adverb obeys the constraints

1 since there is no V-2 in subordinate clauses. Yet the clause final position is
 2 also licensed by these constraints but the result is ungrammatical. A last-
 3 resort type constraint must be imposed to the effect that the adverb must
 4 be linearized in the ‘earliest’ position in which it is licensed. I expect this
 5 constraint on linearization to apply only to weak adverbs which must be
 6 incorporated. The processing burden would be increased if the search for
 7 an appropriate PI position wouldn’t stop after the first hit. Strong adverbs
 8 are not expected to be subject to this constraint, since they are not limited
 9 by the need to incorporate. Whereas the positions of weak adverbs can
 10 depend on the availability of a host, the positions of strong adverbs are
 11 determined by the availability of prosodic boundaries. The constraint
 12 on weak adverb placement may not be too high a price to pay for an
 13 otherwise straightforward account based on independent phonological
 14 properties.

15 Double object constructions provide a test case for this approach to
 16 adverb placement. The Danish data is provided in (26)–(27).

- 17 (26) a. Jeg gav+ikke Peter bogen.
 18 I gave not Peter the book
 19 b. Jeg gav+ham+ikke bogen.
 20 I gave him not the book
 21 c. Jeg gav+ham+den+ikke./*Peter den ikke.
 22 I gave him it not /*Peter it not
 23 d. I går gav+jeg+ham+den+ikke.
 24 Yesterday gave I him it not
 25 (27) a. Jeg gav+ikke bogen til Peter
 26 I gave not the book to Peter
 27 b. Jeg gav+den+ikke til Peter.
 28 I gave it not to Peter
 29 c. Jeg gav+den+ikke til ham/*den ham ikke til
 30 I gave it not to him/* it him not to
 31 d. I går gav+jeg+den+ikke til ham
 32 yesterday gave I it not to him
 33

34 In (26)a the adverb is linearized in the first possible position in which it
 35 can occur. In (26)b the adverb must be placed after the pronoun in order
 36 not to disrupt PI. In (26)c the adverb must be placed after both pronouns
 37 for the same reason. With a non-pronominal indirect object, the adverb

1 must be linearized immediately after the verb, hence OS is blocked. In
 2 (26)d the clitic cluster includes the subject.¹⁶ In (26)b–d the adverb in-
 3 corporates as the final element in the cluster.¹⁷ (27)c shows that the object
 4 cannot ‘shift’ across the particle as expected under the view that objects
 5 don’t shift. Otherwise the examples in (27) parallel those in (26).

6 Holmberg, 1999, (p. 17, (43)) points out that OS is possible in double
 7 object constructions in Swedish when the indirect object is removed:¹⁸

- 8 (28) a. Vem gav du den inte?
 9 Who gave you it not
 10 ‘Who didn’t you give it to?’
 11 b. Henne visar jag den helst inte
 12 her show I it rather not
 13 ‘I’d rather not show it to HER.’

14 This follows without further ado from the restrictions on adverb lineari-
 15 zation outlined here.

16 OS also occurs with pleonastic subjects of small clauses, as pointed out
 17 in Holmberg 1999, a challenge for some syntactic accounts of OS.¹⁹

- 18 (29) a. Jeg hørte *det* ikke regne.
 19 I heard it not rain
 20 ‘I didn’t hear it rain.’
 21

22
 23 ¹⁶ Here I adopt Josefsson’s 1993 idea that object pronouns can form ‘clitic clusters’.

24 ¹⁷ According to Christer Platzak, Swedish also allows (i), but not (ii):

- 25 (i) Han gav den henne inte.
 26 He gave it her not
 27 (ii) *Han har givet den henne
 28 he has given it her

29 The only way I could account for this would be by assuming phonological metathesis of
 30 the two pronouns in the cluster (positioning the longer pronoun after the short one) in
 31 order to ease the further incorporation of the adverb. This explanation would explain
 32 why metathesis doesn’t occur in (ii).

32 ¹⁸ (28)a is ungrammatical in Danish on a par with the English ‘*Who did you give it?’ See
 33 Erteschik-Shir, 1979, for why this is so in these languages. Why Swedish allows ques-
 34 tioning of the indirect object requires further investigation.

34 ¹⁹ Holmberg argues, based on data such as that illustrated in (29), that OS applies to ele-
 35 ments which are marked [–Foc]. Clearly these pleonastics are neither topics nor foci. In
 36 view of the fact that pronouns are necessarily topics, i.e., can be considered to be
 37 [–Foc], Holmberg generalizes OS of topics with the cases in (29) by labelling them all
 [–Foc].

- 1 b. Han tar *den* ikke med ro.
 2 He takes it not with ease.
 3 ‘He doesn’t take it easy.’

4 The requirement on prosodic incorporation of weak pronouns argued for
 5 here, naturally carries over to pleonastics, which are phonologically iden-
 6 tical in that they cannot be stressed. PI is therefore required of pleonastics
 7 as well. Here again the adverb is pronounced in the first position in which
 8 it is licensed. (Note that if the adverb is adjoined to the VP, its domain of
 9 linearization will include the small clause.)

10 The approach to adverb placement suggested here offers an interest-
 11 ing account of the placement of negative QPs in Scandinavian. Sells,
 12 2001 (95–97), following Christensen, 1986, demonstrates that their place-
 13 ment is that of adverbs. The following Danish examples are parallel to
 14 Sells’ Swedish ones:

- 15 (30) a. Hun havde ikke sagt noget.
 16 she had not said anything
 17 b. *Hun havde sagt ingenting
 18 she had said nothing
 19 c. Hun havde ingenting sagt.
 20 she had nothing said
 21

22 (30)b illustrates that even though it is an object, the NegQP cannot occur
 23 in canonical object position. Instead, it occurs before the verb, the canon-
 24 ical adverbial position. This position is also the one in which we find ad-
 25 junct NegQPs as expected:

- 26 (31) Han vil under ingen omstændigheder læse romaner.
 27 he will under no circumstances read novels
 28

29 Accounting for the order in (30) is not simple since the object must merge
 30 VP-internally. If, however, negation like any adverbial is adjoined in a
 31 third dimension and only the nominal part of the phrase is merged in ob-
 32 ject position, we get the following string for (30)c.

- 33 (32) hun havde ingen sagt –ting
 34

35 The first position in which the adverbial can be linearized is after the aux-
 36 iliary as indicated. The object ‘–ting’ is not an independent morpheme
 37 and therefore cannot be pronounced in situ. It is therefore linearized

together with its adverbial host, resulting in (30)c.²⁰ In cases in which the nominal remnant is Ø, as in (33), no phonological displacement of the object is necessary.

(33) Hun havde ingen set.

NegQPs also occur in subject and topicalized positions:

- (34) a. Ingen så mig.
 Noone saw me
 b. Ingenting havde hun sagt.
 nothing had she said

Since the subject position in this case is identical to the adverbial position, they cannot be distinguished. Topicalizing adverbials is generally possible, therefore the occurrence of (34)b is also not surprising.

The placement of NegQPs interacts in an interesting way with OS as shown in (35).

- (35) a. Jeg gav ikke Else noget.
 I gave not Else anything
 b. *Jeg gav Else ingenting
 I gave Else nothing
 c. *Jeg gav ingenting Else
 d. Jeg gav hende ingenting.
 I gave her nothing

(35)a shows that the order of objects is IO followed by DO. (32)b is ruled out because the adverb must be linearized earlier. (32)c is ruled out by ID (extended to include both direct and indirect objects: the first object is IDed as the indirect one unless otherwise marked). (32)d, with an IO pronoun is pronounceable because in this case the final position is the first position in which the adverbial can be linearized. The NegQP data in Scandinavian therefore falls out naturally from the three-dimensional analysis of adverbials together with the free linearization approach outlined here.

²⁰ As pointed out to me by Lisa Rochman, this analysis is identical to *neg*-incorporation into indefinites in Klima, 1964, 274. There, however, the fusion was syntactic.

Section 2 started out with a free-ride approach to OS and ended up with free linearization of adverbs.²¹ Linearization of adverbs is phonologically constrained (by requirements on PI and by ID), however there is no longer any phonological displacement.

2.4. *The phonology of prosodic incorporation*

PI is a phonological process which joins elements of a phonological string to make one prosodic unit. This prosodic unit allows for no prosodic breaks to intervene between its elements and therefore cannot be broken up by the insertion of lexical elements such as adjuncts. These prosodic units make up prosodic words. Potential cases of PI include clitics and their hosts, cases of contraction (e.g., auxiliary and negation in English) and compounds. Selkirk, 1996 offers an account of the prosody of functional category words and argues that they may appear in a variety of prosodic clitic structures and that they need not have the status of prosodic words. Her analysis of weak object pronouns is particularly pertinent here.²² According to Selkirk, weak object pronouns are affixal clitics which do not themselves constitute a prosodic word. I assume the same is true of Scandinavian object pronouns. Following Selkirk, the prosodic structure of the V+object of (1) is as shown in (36) (parentheses indicate prosodic words; the pronouns are presented in their reduced incorporated form).

²¹ It remains to be seen whether this approach to adverb placement can handle the mass of related insights, prominent among them Cinque, 1999 and, for Scandinavian, Nilsen, 2003.

²² Selkirk argues that strong and weak object pronouns have different syntactic structures. Weak pronouns form a constituent with the verb, whereas strong pronouns occur in the normal object position. She notes (p. 204, fn. 9) that pronoun clusters (of double object constructions) are sisters in prosodic structure. Selkirk does not offer an account of these, yet her approach requires that such clusters together form a syntactic constituent with the V, a serious problem with the syntactic account. Selkirk also offers an account in terms of optional prosodization of the two types of pronoun. This approach, however, requires that functional categories are referred to by prosodic rules, which goes against the basic claim of the article. In view of the fact that strong pronouns are foci whereas weak ones are topics, it is possible to employ these properties to account for the different prosodic properties of the two types of pronouns. In section 6.1, I discuss the place and role of topic/focus marking in the grammar.

- 1 (36) ((mødte) 'am)
2 met him

3 This analysis can be extended to all weak pronouns.

- 4
5 (37) a. (ja' ((mødte) 'am)) or (ja' (mødte)) 'am)
6 I met him
7 b. (ja' (mødte))Peter
8 c. Peter ((mødte) 'am)
9 d. ((ga) 'am d'n)
10 gave him it

11 (37)a shows that a subject pronoun incorporates into the prosodic word
12 as well. In principle there are two bracketing options, both of them in-
13 dependently necessary, as shown in (37)b and (37)c. It follows that there
14 is no way to choose between the two options in (37)a. I conclude that
15 both pronouns incorporate directly into the verb in the same way double
16 object pronouns do in the 'clitic cluster' in (37)d. I therefore assume the
17 flat prosodic structure for the incorporation of multiple pronouns as
18 shown in (38).²³

- 19 (38) a. (ja' +(mødte)+'am)
20 I met him
21 b. ((ga)+'am+d'n)
22 gave him it
23

24 Word stress rules will ignore elements that are not prosodic words within
25 these strings so that only the verb itself is stressed and the weak pronouns
26 are pronounced as stressless syllables. Weak pronouns can also incorpo-
27 rate into DPs:

- 28 (39) Sara gav Peter+d'n.
29 Sara gave Peter it
30

31 Incorporation of weak pronouns into DPs is not possible in English
32 which is why the translation of (39) cannot be pronounced. It is pre-
33 dicted that language variation will often depend on language particular
34

35 ²³ I do not take a stand here on whether hierarchical structure is necessary on the prosodic
36 phrase or intonational phrase levels as argued by Selkirk, 1984 and Nespor and Vogel,
37 1986 among others.

constraints on PI, such as which constituent types can provide hosts and which elements must PI. A careful analysis of the phonological properties of hosts, on the one hand, and elements which PI, on the other, might shed light on the source of these differences.

In section 2.2, I showed that Scandinavian adverbials also occur in strong and weak forms and that the latter incorporate. Most adverbials have at least two syllables which means that they must be analyzed as prosodic words and can in principle receive stress even when they are incorporated. Their incorporation therefore differs from that of weak pronouns as shown in (40). Here two prosodic words are combined by PI to form one complex prosodic word:²⁴

(40) ((mødte)+(ikke))

Weak adverbs, like weak pronouns, can incorporate both to the left (enclitic) and to the right (proclitic) as shown in the examples in (41).

- (41) a. Peter mødte+ikke Sara
Peter met not Sara
b. at Peter ikke+mødte Sara.
that Peter not met Sara
c. Peter har ikke+mødt Sara.
Peter has not met Sara

Pronouncing a pause at the site of PI leads to ungrammaticality. I take this as evidence for the correctness of the attachments indicated in (41). Adverbs also PI in clusters:

- (42) Peter mødte+muligvis+ikke Sara.
Peter met possibly not Sara
'Peter possibly did not meet Sara.'

The order of the adverbs in the cluster is determined both by attachment site and by other phonological properties of the individual adverbs. In (42) *muligvis* is a sentential adverb with a higher attachment site than the

²⁴ If adverbials are lexical, then this analysis would not follow from Selkirk's approach. It is possible that reasons can be found not to treat the result of adverb incorporation as a prosodic word, but rather a prosodic phrase. I do not take a stand on this here. What is important to me is only that PI must occur.

VP adverb *ikke*, hence the first possible attachment site of the former precedes that of the latter and these adverbs cannot be reordered.²⁵

If both pronouns and adverbs form clusters, then it is no longer obvious why weak pronouns cannot follow adverbs in an incorporated cluster, leading to the ungrammatical (43).

- (43) *Peter mødte+ikke+ham
 Peter met not him

There is actually a good reason for this. Compare the two sequences in (44):

- (44) a. (mødte)+ham+(ikke)
 b. (mødte)+(ikke)+ham

The difference between (44)a and (44)b lies in the different phonological properties of the incorporated pronoun and the incorporated adverb. The former is treated phonologically as a stressless final syllable of its host and is therefore invisible to further incorporation. The latter is itself a stressed prosodic word, therefore visible to incorporation. Since it is weak, it is not a possible host for further PI, ruling out (44)b. In the next section I discuss the reasons why the order in (44)b is in fact possible in Swedish.

An interesting upshot of this view of the pronunciation of adverbs provides a potential explanation for the different distribution of *not* and other adverbials in English. Let us hypothesize that English adverbials differ from the Scandinavian ones as follows: 1. *not* is unique among English adverbs in having only a weak form which only optionally forms a prosodic word. 2. other English adverbials have only a strong form.²⁶ The following data falls out, without requiring an additional syntactic distinction between Negation and adverbials as is customary:

- (45) a. Peter did+not/didn't meet Sara.
 b. *Peter did obviously not meet Sara.

²⁵ Erteschik-Shir and Rapoport, in preparation, propose that verbs are decomposed into meaning components with associated syntactic projections. The attachment sites of adverbs will be constrained by the meaning component modified, leading to different attachment sites with scopal consequences.

²⁶ Prosodic properties of lexical elements are to a large extent parametric, yet individual lexical items may have exceptional properties for which they must be marked.

- 1 (46) a. Peter never met Sara.
 2 b. *Peter met never Sara.

3 (45)a and (46)a show that English VP adverbials linearize before the VP.
 4 This is possible in English which does not require V-2. (45)b shows that
 5 nothing can intervene between *not* and its host, evidence that *not* is neces-
 6 sarily weak and therefore must PI.²⁷ Its host cannot be the subject since
 7 this would position the adverb outside the VP domain. *Do*-support occurs
 8 to provide a verbal host and is therefore taken to be prosodically moti-
 9 vated.²⁸ The position of the adverb in (46)a is the first position in which
 10 the adverb can be linearized, but not the only one, since, as I suggested in
 11 the previous section, strong adverbs can occur at any prosodic boundary
 12 within their domain. Since verb phrases form prosodic phrases, adverbs
 13 cannot occur inside them, blocking (46)b. Such sequences are generally
 14 thought to be blocked by some syntactic implementation of an adjacency
 15 requirement on case assignment. This requirement would have to hold in
 16 English but not in Scandinavian, yet there is no obvious explanation
 17 for this difference. The prosodic explanation follows from the difference
 18 in V-2. We return to a discussion of V-2 and its motivation in section 6.2.

19 Costa, 2004 also observes that light adverbs behave in a clitic-like
 20 fashion and therefore argues that they are syntactic heads. In this way he
 21 can account for the fact that adverb position depends on weight. In my
 22 account, no syntactic distinctions need to be made between light and
 23 heavy adverbs or pronouns. Syntactically they are uniform, phonologi-
 24 cally they differ.

25 PI has been employed here to explain constraints on adverb place-
 26 ment in Scandinavian, raising new questions about prosodic properties
 27 of adverbs in different languages. Further research should unveil what
 28

29
 30 ²⁷ (i) shows that a strong (contrastive) form of *not* occurs in questions in which *do*-support
 31 is triggered by inversion and not by the negative element itself.

32 (i) Did Peter *nót* see Mary?

33 This example does not refute my claim that *do*-support with negation stems from nega-
 34 tion being weak, but it does show that the phonology of *do*-support requires a more
 35 detailed investigation.

36 ²⁸ In view of their different prosodic structure the two forms are also stressed differently:

37 i. ((*did*)+*n't*)
 ii. ((*did*)+(*nót*))

morpho-phonological properties determine properties of PI in a particular language. What, for example, is the explanation for the different prosodic properties of adverbs in Scandinavian and English, and why do certain constituent types qualify as hosts of PI in one language but not in another. These differences should be made to follow from other language particular morpho-phonological properties. PI may also provide an explanation for more phenomena which have until now been believed to be syntactic.

3. Variation

It has often been observed (e.g., Holmberg, 1986; Josefsson, 1992, Sells, 2001), that in Swedish, object shift is optional. According to Sofie Raviv (personal communication) OS is also not required across negation in the *Ærø*/South Fyn dialect of Danish. The examples in (47) illustrate weak object pronouns following negation.

- (47) a. Jag såg +inte+den. (*Swedish*)
 I saw not it
 b. Jeg så +ikke+det (*Danish Ærø/South Fyn dialect*)
 I saw not it

Anders Holmberg (personal communication) suggests that the adverb and the pronoun in (47)a are pronounced as a clitic cluster with both PI-ed.²⁹ Since both the adverb and the pronoun must PI, and neither can host the other, this would be the only way to pronounce the sequence. Yet in the previous section I argued that in Danish (except for the *Ærø* dialect) this is not possible. Swedish differs from Danish in allowing phonological reanalysis of the verb with the incorporated adverb, with the stress on the adverb perceived as word stress on the verb+adverb sequence. In this case, then, it is the verb+adverb together which provides a host for the pronoun, not the adverb alone. Holmberg also provided me with the following Swedish example which can be viewed as evidence for this analysis (see also Holmberg 1999, 28 for the same point in Norwegian):

- (48) ?Han såg utan tvivel den.
 he saw without doubt it

²⁹ See Holmberg, 1984 and Josefsson, 2001 for an analysis of Swedish particles as clitics.

1 According to Holmberg acceptability is reduced with a more
2 phonologically-complex adverb. It is well known that all things being
3 equal, complex phonological elements are preferred after simple ones
4 when possible.³⁰ Since the adverb is licensed both before and after the
5 pronoun, the order which is most easy to pronounce, that with the
6 heavier adverb at the end, will be preferred. (In the *Ærø* dialect the phe-
7 nomenon is limited to negation.)

8 The Danish example in (35)b also corroborates this analysis. According
9 to Sofie Raviv (pers. com.), the *Ærø* dialect is spoken with a special tune
10 which makes it sound like Swedish and (35)b is pronounced with main
11 stress on the incorporated negative adverb. This pronunciation of nega-
12 tion is completely ruled out in standard Danish. Stress shift onto the
13 incorporated adverbial is therefore the trigger for licensing the further
14 incorporation of the object pronoun in situ. Another piece of evidence
15 for this analysis is the following Swedish example (Elisabet Engdahl, per-
16 sonal communication):

- 17 (49) a. Där vill aldrig+nån bo.
18 there wants never someone live
19 ‘No one wants to live there ever.’
20 b. Där vill nån aldrig bo.
21 there wants someone never live
22 ‘A certain person wants to never live there.’

23 The order in (49)a is possible in Swedish on a par with (47)a. The pro-
24 form ‘nån’ can incorporate into the adverb when the adverb is incor-
25 porated into the verb and stressed. (49)b shows the shift of the pronoun,
26 here a subject pronoun. Note that the order of the pronoun and the
27 adverb determines the scopal relation. In (49)a the pronoun is nonspecific,
28 in (49)b in preadverbial position it must be specific. (I return to this topic
29 in section 6.1). Neither of the examples in (49) are grammatical in Danish.
30 (49)a is bad because pronouns cannot PI into verb-adverbial sequences in
31 Danish. (49)b is not possible because the parallel form in Danish *nogen*
32 cannot be used specifically.³¹

34 ³⁰ An obvious example is heavy NP shift. See Ross, 1975 for further examples.

35 ³¹ Christensen, 2003 offers an OT analysis of dialectal variations in NegQP shift. I am
36 optimistic that a careful analysis of cross-linguistic phonological differences will shed
37 some light on the reasons for the various constraint rankings.

The possibility of incorporating both an adverb and a pronoun into the verb in Swedish also allows for the word order in (50)a (Halldór Sigurðsson, personal communication):

- (50) a. Det vet+inte+jag.
that know not I
'That I don't know.'
b. Det vet+jag+inte.
that know I not

Here the object is topicalized and it is the subject which undergoes PI into the verb either before or after the adverbial.

The possibility of 'adverbial intermingling' in Swedish was noted in Holmberg and Platzack, 1995 (156–7). The examples in (51) are from Sells, 2001 (63, his (121)):

- (51) a. Igår läste han dem ju alltså troligen inte.
Yesterday read he them as.you.know. thus probably not
'Yesterday as you know he probably did not read them thus.'
b. %Igår läste han ju dem alltså troligen inte.
c. %Igår läste han ju alltså dem troligen inte.
d. %Igår läste han ju alltså troligen dem inte.
e. %Igår läste han ju alltså troligen inte dem.

(51)a is the normal case of OS with a string of adverbs following the object. In the rest of the examples, the pronoun follows one or more of the adverbs. This is possible only in Swedish and, as in (48), a string of adverbs preceding the object makes it less easy for the adverbs to incorporate. The oddness is therefore due to difficulty of pronunciation. This is also the explanation for the oddness of (51)b–d. These are all possible in Swedish since adverbs are not blocked from being linearized either before or after the object pronoun, yet the incorporation of a longer string of adverbs and pronouns is problematic. Holmberg and Platzack employed this data to argue against a cliticization analysis in which the pronoun cliticizes onto the verb, very much along the lines of the current approach. However, together with the free linearization of adverbs, the analysis is still viable.

'Long Object Shift' is another phenomenon possible only in Swedish (Josefsson, 1992). The examples in (52) are from Sells, 2001 (65, his (124)–(125)):

- 1 (52) a. Numera gör mej alltså inte längre Helge lika irriterad
 2 Nowadays makes me thus not longer Helge as irritated
 3 som förr.
 4 as before
 5 ‘So nowadays Helge doesn’t make me as irritated as before
 6 any longer.’
 7 b. Numera gör alltså mej inte längre Helge lika irriterad som
 8 förr.
 9 c. Numera gör alltså inte mej längre Helge lika irriterad som
 10 förr.
 11 d. Numera gör alltså inte längre mej Helge lika irriterad som
 12 förr.

13 According to Sells, the acceptability of (52)b and (52)c is reduced. This
 14 would follow from the same type of considerations as those pertaining to
 15 (51) (which would also predict the reduced acceptability of (52)d). More
 16 seriously, long OS violates the ID constraint in that the object pronoun
 17 intervenes between the verb and the subject (see (16) in section 2.2).
 18 Josefsson, 1992 considers constructions such as these highly marked and
 19 lists restrictions of various kinds on their occurrence. First of all, Long
 20 OS is restricted to cases in which the subjects are of the type which would
 21 have received quirky case in Old Swedish. If quirky case is thematically
 22 motivated, then these subjects can be IDed by their thematic role thus
 23 relaxing the prohibition on an intervening object. Another requirement is
 24 that the elements around the object pronoun must be ‘heavy’, this accord-
 25 ing to Josefsson to avoid stress on the pronoun, clearly a phonological re-
 26 striction. Finally, the morphology of the pronoun is also important. Only
 27 object pronouns with unambiguous overt case marking are licensed in
 28 this position, clearly consistent with a processing constraint such as ID.
 29 The following example provided by Halldór Sigurðsson and Elisabet
 30 Engdahl illustrates this:

- 31 (53) I går träffade honom hans föräldrar på stationen.
 32 yesterday met him his parents on the station
 33 ‘Yesterday his parents met him at the station.’
 34

35 If the pronoun *honom* is replaced by the ambiguous *dom* (‘they’ or
 36 ‘them’), then the only available reading is that in which the pronoun is
 37 interpreted as a subject, exactly as predicted by ID.

Although a counterexample to the ID ban on elements with $\varnothing$ -features, Long OS provides strong evidence for the phonological account of OS, in which both processing and pronunciation play a role. Yet it is not obvious how this order is derived. No obvious syntactic analysis presents itself. In view of the requirement of heaviness on either the subject or the other surrounding elements, I can only suggest that the subject is moved (to the right) beyond the object pronoun by a type of phonological meta-thesis akin to heavy NP shift.

Yet another difference between Swedish and the other Scandinavian languages is that OS is obligatorily blocked across particles in Swedish.

- (54) a. Vi kastade+inte+ut+den/*den inte ut. (*Swedish*)
 we threw not out it
 b. Vi smed den ikke ud/*ikke ud den. (*Danish*)
 we threw it not out

This difference follows from the difference in particle object order in these languages according to Sells, 2001, 48 as illustrated in (55):

- (55) a. Vi kastade ut boken /*boken ut. (*Swedish*)
 We threw out the book/*the book out
 b. Vi smed bogen ud /*ud bogen. (*Danish*)
 we threw the book out/*out the book

Sells, p. 68, following Holmberg, 1986, assumes that the verb-particle complex forms a unit dominated by V in Swedish. It follows that the object cannot follow the verb but must follow the whole complex.³² This complex cannot be formed in Danish. I again assume that the reason for this has to do with the possibility of pronouncing this complex in the two languages. In Swedish the incorporated particle is stressed in the same way the incorporated negative adverb is in (47)a. In Danish only phrase final particles can be stressed. This prosodic pattern may be a requirement on particle incorporation. Once this pattern is available, the pronoun can also be incorporated, again parallel to (47)a. In Swedish it is the Verb+adverb or Verb+part complexes which host PI.

³² The data is actually somewhat more complex. Here I deal only with one type of particle and ignore particles with complements. Note also that Norwegian and Icelandic allow both orders in (55), yet OS is as in Danish (see Josefsson, 1992 and Bobaljik, 2002).

The placement of the adverb must still be explained. (56) gives the three possibilities of which (56)a is the desired result:

- (56) a. vi kastade **inte** ut den
 b. *vi kastade ut **inte** den
 c. *vi kastade ut den **inte**

(56)a is pronounceable since the adverb PIs into the verb and the particle and the object pronoun follow suit. The unmarked pronunciation of the sequence has stress on the particle and not on the adverb (stress on the adverb is contrastive). This renders the appropriate contour for further incorporation of the pronoun. (56)b and c are ruled out by the last resort constraint introduced in 2.3, namely that weak adverbs must be linearized in the 'earliest' position in which they are licensed. A more detailed investigation of Swedish phonology may provide additional reasons.³³ The pronoun-adverb-particle order in Danish (and the other Scandinavian languages) also follows:

- (57) a. *vi smed **ikke** den ud
 b. vi smed den **ikke** ud
 c. *vi smed den ud **ikke**

In (57)a the pronoun cannot PI since Danish adverbials are not stressed the way they are in Swedish and so the V-adverb unit does not provide a host for PI, thus leaving the pronoun stranded. (57)b allows for the pronoun and adverb to PI and is therefore licensed. In (57)c the particle cannot host the incorporation of the adverb and it also violates the last resort principle.

Bobaljik, 2002 relates the lack of OS across particles in Swedish to analytic passive formation. (236, his (51a)):

- (58) *Skräpet måste bli kastat ut
 the scrap must Aux thrown out
 'The scrap had to be thrown out.'

In contrast the analytic passive is possible in Danish, Norwegian, and Icelandic. (59) illustrates the parallel example in Danish.

³³ According to Árnason 1996, Danish is exceptional among the Scandinavian languages in not adhering to Prokosch's Law (i.e., that all stressed syllables are heavy). Danish therefore allows short stressed syllables. This may also turn out to be a source of differences in PI.

- (59) Affaldet blev smidt ud.
 the scrap Aux thrown out
 'The scrap was thrown out.'

Bobaljik draws the following generalization: "An object which can only appear in the post-particle position is blocked from further A-movement (i.e., object shift and passive).³⁴ This generalization would follow simply from the assumption that the particles in question cannot incorporate phrase-finally in Swedish. In Svenonius, 1996, evidence both for and against this suggestion can be found. Svenonius shows that particles incorporate obligatorily in the analytic passive in Swedish, exactly in those cases in which they cannot occur in final position (his (49)):

- (60) Mjölken blev aldrig inställd/*ställd i
 the milk was never in.put put in
 'The milk never got put away'

Unfortunately, the Swedish synthetic passive allows a final particle (Svenonius' (66)):³⁵

- (61) a. Hunden kastades ut.
 the dog threw-pass out
 'The dog was thrown out'
 b. ?? Hunden utkastades.
 the dog out.threw-pass

The distribution of sentence final particles in Swedish also shows up with Neg QPs. (62) is from Sells, 2001 (98) (used by Sells to discuss the position of the NegQP):

- (62) a. ?Jon har inga romaner läst ut.
 John has no novels read out
 b. *Jon läste inga romaner ut.
 John read no novels out

³⁴ Bobaljik, 2002 proposes an account in terms of locality of A-movement rather than the one proposed in Holmberg, 1999 in which examples such as the unacceptable string in (54)a violate the adjacency requirement between the base and shifted position of the object. See Section 5 for a comparison of Bobaljik's account with the one proposed here.

³⁵ Svenonius argues for a correlation across the Scandinavian languages between A-movement, participle agreement and particle incorporation. A discussion of this and the rich data he supplies would take us too far afield.

1 Here the aux-participle in (62)a is much more acceptable than the finite
2 verb form in (62)b. The two kinds of passive also differ in this way, but
3 the final particle is blocked in the analytic one. I expect that with more
4 investigation into the phonology of these constructions, an account of
5 the distribution of final particles in Swedish will be uncovered.

6 The ideas presented in this section are somewhat speculative. Yet, the
7 strength of a phonological explanation for fine distinctions between other-
8 wise highly similar languages is that it allows us to maintain the same
9 syntax for these languages by accounting for variation in terms of mor-
10 phological and phonological distinctions. Optimality theory started out
11 as a theory of phonology. More recently, it has been applied to syntax
12 and in particular to explain variation (e.g., Sells, 2001 who offers an OT
13 account of OS). If variation is indeed to a large extent phonologically
14 determined by language specific phonology and morphology, the success
15 of OT in this area is not surprising. What is missing in OT accounts, in
16 my opinion, is the association of the syntactic distinctions with the actual
17 phonology and morphology of the languages in question.

18

19

20 **4. Full DP OS in Icelandic**

21

22

23 The free-ride approach to OS required an analysis in which DP objects
24 could PI into a verbal host, not a very natural phonological process.
25 Free linearization of adverbs leads to a different explanation of the dif-
26 ference between Icelandic and the other Scandinavian languages. An ob-
27 vious difference is the fact that Icelandic has morphological case. Since
28 overt morphological case interacts with ID, a potential explanation
29 for the difference presents itself, if it weren't for the fact that Faroese
30 also has overt case but doesn't allow OS of full DPs.³⁶ Here I pursue an

31

32

33 ³⁶ Holmberg and Platzack, 1995, 172–176 argue that Faroese morphological case is
34 weaker than that of Icelandic thus explaining the difference in OS. Bobaljik and
35 Thráinsson 1998 argue that the presence of agreement phrases is indirectly linked to the
36 availability of overt agreement morphology. They argue that the difference between
37 Faroese and Icelandic is a difference in inflectional morphology and does not have to
do with the availability of overt case.

examination of phonological differences between Icelandic and Faroese that predicts that only in Icelandic can the sequence DP+adverb be pronounced.

(63)a and b illustrate the optional OS of a full (definite) DP in Icelandic.

- (63) a. Jón las bókina ekki.
Jon read the books not
b. Jón las ekki bókina.
Jon read not the books

These sentences were uttered by a speaker of Icelandic, rendering the pitch contours (without the subjects) in figures 1 and 2:

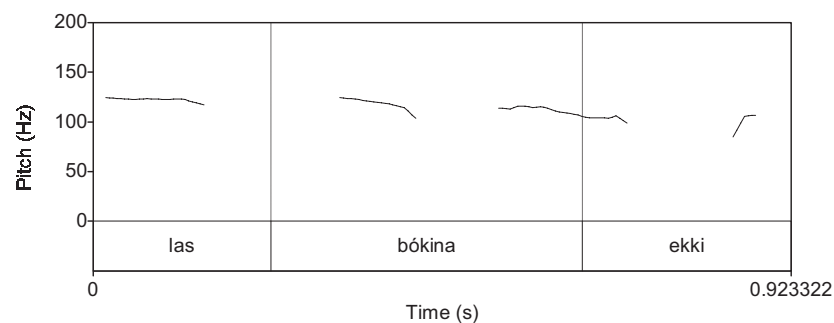


Figure 1. F_0 of (63)a

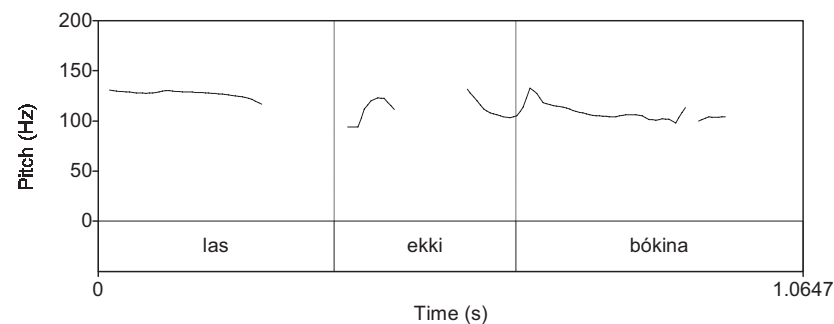


Figure 2. F_0 of (63)b

1 In neither case is there evidence of PI into the verb. However, the nega-
 2 tive adverb and the object form a prosodic unit in both orders. This is as
 3 clearly heard as is the lack of incorporation into the verb. Assume, there-
 4 fore, that Icelandic adverbs, like all Scandinavian adverbs, are weak and
 5 must PI. Yet whereas Danish adverbs PI into the verb, Icelandic adverbs
 6 PI into the object (either pro- or enclitically) in these cases:

- 7 (64) a. 'bókina+, ekki
 8 books-the not
 9 b. 'ekki+, bókina³⁷
 10

11 As expected heavy noun phrases do not shift:

- 12 (65) a. Ég sá ekki ráðherrann sem allir eru að kvarta yfir.
 13 I saw not minister-the who all are to complain over
 14 'I didn't see the minister who everyone was complaining
 15 about.'
 16 b. ??Ég sá ráðherrann sem allir eru að kvarta yfir ekki.
 17 I saw minister-the who all are to complain over not
 18

19 This follows naturally from an account in terms of prosodic incorpora-
 20 tion.

21 According to Árnason 1996, Icelandic and Faroese are unique among
 22 modern Western European languages in having initial stress in most of
 23 their vocabularies. Having initial stress might enable a language to toler-
 24 ate a longer sequence of syllables without primary stress than a language
 25 with primary stress further towards the end of the word, thus allowing
 26 for longer prosodically incorporated strings. This reasoning should also
 27 apply to Faroese. Icelandic and Faroese differ, however, in the way they
 28 form compounds. In Icelandic, stress generally remains initial. In Faro-
 29 ese, stress generally shifts to the second element of the compound in bor-
 30 rowed compounds.³⁸ The fact that Faroese shifts the stress to the second
 31 element of the compound in new compounds indicates that this shift is
 32 productive in Faroese. A possibility worth considering is that (64)a is
 33

34
 35 ³⁷ It is in fact hard to detect a difference in stress between the two. This may be due to the
 36 fact that phrasal stress is final.

37 ³⁸ Such compound stress generally mimics the Danish stress pattern in Danish borrowings.
 See Árnason for details.

1 ruled out in Faroese because it would be unpronounceable with stress
2 shifted onto the incorporated adverbial.³⁹

3 Icelandic differs from MSc in that DPs can host PI of adverbs. This
4 may also be the source of the difference between Icelandic and Faroese.
5 If DPs cannot host adverbial PI, then (64)a would be ruled out. Section
6 2.4 showed that differences in potential PI hosts can explain word order
7 differences across languages. Further research into the phonology of
8 both Icelandic and Faroese is required to discover which of these explanations
9 (if any) is tenable.

10 Imagine a language with the stress properties of Icelandic but without
11 overt case. I predict that full DP object shift would not be possible in such
12 a language because ID would be violated. Overt case marking enables the
13 identification of the object independently of its position with respect to
14 the adverb.

15 If both orders are pronounceable and neither functions to ID the ob-
16 ject, what then motivates Icelandic object shift? It is here that the inter-
17 pretation of the shifted DP comes into play. It is well known that the
18 shifted DP must be a topic (see Diesing and Jelinek, 1993, among others).
19 Only definites shift and shifted bare plurals must be specific.⁴⁰ Shifting the
20 object thus identifies it as a topic. The ID constraint as defined in (16) is
21 concerned with the identification of the syntactic roles of arguments. Yet
22 the identification of topic and focus is also critical to the interpretation of
23 the sentence. The identification of foci is through intonation, the identifi-
24 cation of topics is through fronting. An element fronted to the initial
25 position in the sentence is identified as the main topic of the sentence, an
26 element fronted within the VP is interpreted as a subordinate topic.⁴¹ An-
27 other way to mark subordinate object topics is by not stressing them. This
28 is the strategy used in the other Scandinavian languages and in English.
29 In other Germanic languages such as German and Dutch, topics undergo
30 Scrambling. Scrambling has much in common with Scandinavian OS, yet
31
32

33 ³⁹ Another possibility is that just as Danish pronunciation affects compound stress in
34 Faroese (see Árnason 1996), it may also directly affect Faroese adverb linearization.

35 ⁴⁰ See Cohen and Erteschik-Shir, 2002 for an analysis of specific bare plurals as topics and
36 Erteschik-Shir, 2001; Erteschik-Shir and Strahov, 2004 for arguments that shifted ob-
37 jects in Icelandic must be topics.

⁴¹ See Erteschik-Shir, 1997 for properties of subordinate topics.

it occurs independently of verb movement. An examination of scrambling along the lines outlined here for OS should be fruitful.

5. A comparison with Bobaljik, 2002

Bobaljik's 2002 account is an ingenious way to allow for a purely syntactic account of OS, sensitive to phonology (adjacency), without PF filters on syntactic derivation. Since Bobaljik's proposal, in principle, could incorporate the phonological restrictions on OS proposed here, the differences between the two approaches shed light on the strengths and weaknesses of both.

Bobaljik's core idea is that LF and PF have the ability to privilege either the higher or lower position in the chain, allowing a moved element to be pronounced (and interpreted) as though it had not moved. His account of OS is as follows: All objects undergo OS to spec, AgrOP. Phonological considerations determine which copy is pronounced and semantic considerations which copy is interpreted. When verb movement does not occur, only the lower copy of the object can be pronounced, since the position of the higher copy intervenes between Infl and the verb stem blocking their PF-/morphological merger which requires adjacency.⁴² This is illustrated in (66) (Bobaljik's Icelandic example (40), p. 225).

- (66) * [_{CP/IP} Hann hefur [_{PartP} – ið [_{AgrOP} **bókina**_i [_{VP} les- ~~bókina~~_i]]]]
He has PRT read
 [Adjacency disrupted: O - - - - - * - - - - - O]

In (66) the higher copy of the object in Spec, AgrOP cannot be pronounced since it disrupts the adjacency between the main verb stem and the projection hosting participial morphology. Had the verb moved (in a non compound tense), adjacency would not be disrupted and the object

⁴² Bobaljik rejects the syntactic account of inflection which requires syntactic raising of the verb to I in these cases. Instead he proposes post-syntactic merger of the finite verb and the inflectional features in Infl.

could be pronounced in the higher position.⁴³ Bobaljik's proposal also provides an explanation of the semantic effect of OS. According to Diesing, 1992, the object is interpreted as new information in the lower VP internal position, and as old information in the higher position. Bobaljik, (p. 251) proposes an economy condition which prefers isomorphism between PF and LF. It follows that when movement is possible (i.e., when the verb has moved), the object will be interpreted in the position in which it is pronounced.⁴⁴

Bobaljik's analysis requires that adverbs be invisible for the computation of adjacency. He achieves this effect by making a set of assumptions about the linearization of syntactic structures. Syntactic structures, according to Bobaljik (213–221), contain only hierarchical information. Linear order is determined at Spell-Out by establishing precedence among syntactic nodes. Adverbs are taken to be adjoined to VP and are therefore not dominated by VP, but only by a segment of VP. If a precedence condition requires that X precede VP, then X must precede the set of terminal nodes dominated by VP, excluding the adjoined adverb which can therefore float to the left of X in the mapping to phonology. The special status of adjunction thus follows from the segment-category distinction. I believe it to be significant that the segment-category distinction is employed by Truckenbrodt, 1999 to account for the mapping from syntax to phonological phrases for exactly the same case. He argues that in the case of adjunction to VP, the higher segment of VP that includes the adjunct is irrelevant to the syntax-prosody mapping, forcing a phonological phrase boundary between the lower VP and the adjunct. Note that if adjuncts are merged in a different dimension the segment-category distinction is not required to account for the linear position of adverbs (as in Bobaljik) nor is it required to constrain the mapping from syntax to phonological phrases (since there is no higher VP, only the lower one will

⁴³ The adjacency requirement between the verb and its inflection is immaterial to my account of OS. My account of adverb linearization, however, does predict that adverbs do not block morphological merger between the verb and Infl. This is due to the fact that linearization of adverbs presupposes lexical insertion (of both of adverbs and verbs) and the inflected verb forms must therefore be formed before adverb linearization.

⁴⁴ It is not clear to me how Bobaljik's approach can be extended to genuinely optional movement such as Topicalization, since topics can be interpreted in situ as well.

count). Instead, phonological phrasing will directly determine the possible positions of unincorporated adverbs.

There are other cases where Bobaljik must resort to rather complex considerations, diminishing the beauty of his basic idea. One illustration is the case of participle fronting which he discusses on p. 234–235. The following example of Stylistic Fronting in Swedish is from Holmberg, 1999, p. 7:

- (67) Kysst har jag henne inte.
Kissed have I her not
'Kissed her, I haven't.'

The problem is that although the verb has moved, adjacency between the verb stem and the participial morphology is disrupted. Bobaljik follows Holmberg in assuming that the participle is fronted by long head movement which "has the effect that adjacency between the head of PartP and the lexical verb stem is not at issue and object shift should be possible." (p. 235). In order to see this let us examine the structure Bobaljik would assign (along the lines of Bobaljik's account of similar cases, e.g., (66)):

- (68) [CP/IP Kyss- har jag [PartP – t [AgrOP **henne**_i [VP **kyss-** **henne**_i]]]
kissed have I PRT her
a. [Adjacency disrupted: O - - - - * - - - - -O]
b. [O - - - - * - - - - -O]

(68)a shows that the object pronoun disrupts adjacency between the participial morphology and the lower copy of the verb and (68)b shows that the subject pronoun disrupts adjacency between the higher copy of the verb and the participial morphology. Either way, merger of the verb and its morphology is blocked. The only way for Bobaljik to explain the grammaticality of the sentence is to assume that the lower copy of the object is pronounced. But then he cannot explain the position of the adverb after the object. (He suggests that in fact the pronoun can (marginally) be pronounced after the adverb, but clearly that is only one option.) This type of example does not present a problem for the current view if it is assumed that the stress on the fronted participle prevents PI of the auxiliary and the adverb so that the adverb cannot linearize after the auxiliary. The first possible position for the adverb is therefore after the object. As

seen in section 3, Swedish allows incorporation of the adverb into the verb before the object. This is why both orders are predicted to be good.⁴⁵

One of the strengths of Bobaljik's approach is his account of the difference between VO and OV languages. The fact that OS is not dependent on verb movement in OV languages follows without further stipulation. The main weakness of his account, in my opinion, is that he does not account for differences among the Scandinavian languages, which are all VO. He admits that his account sheds no new light on why OS is limited to weak pronouns except in Icelandic (beyond Bobaljik and Thráinsson 1998 mentioned in the previous section). More importantly, his approach cannot, as far as I can tell, advance the understanding of more minor differences between the Scandinavian languages such as the fact that OS is optional in Swedish, but obligatory in Danish.

6. Architecture and motivation for phonological displacement

6.1. *Topic-focus architecture*

In previous work (e.g., Erteschik-Shir, 1997) I argued for an architecture without LF-syntax, in which f(ocus)-structure, the output of syntactic computation with constituents marked for topic and focus, provides the input to interpretation.⁴⁶ In this model, f-structure is therefore responsible for scopal interpretation. Since truth value is determined by evaluating the sentence with respect to the topic, topics will always take wide scope.⁴⁷ I repeat the examples in (49) from section 3 as an illustration.

- (69) a. Där vill aldrig+nån_{foc} bo.
 there wants never someone live
 'No one wants to live there ever.'

⁴⁵ As pointed out to me by Idan Landau, Bobaljik's account also fails to account for OS in V-2 double object constructions. As shown in (26) vs. (28), OS is not possible 'across' a full DP indirect object, yet is possible when the indirect object is removed.

⁴⁶ In Erteschik-Shir, 2003, submitted, I show that top and foc features can be lexically selected on a par with ϕ -features. In earlier work, I assumed, in violation of the inclusiveness principle, that syntactic structures are annotated freely with these features and that these are the only features which can be inserted in the derivation.

⁴⁷ This view of topics stems from Reinhart, 1981 and is developed in Erteschik-Shir, 1997.

- 1 b. Där vill nån_{top} aldrig bo.
2 there wants someone never live
3 ‘A certain person wants to never live there.’

4 In (69)a the post-adverbial quantifier is focused and hence has narrow
5 scope, in (69)b the adverbial precedes the adverb, is interpreted as a topic
6 and takes wide scope.

7 The fact that foci are marked intonationally indicates that f-structure is
8 also spelled out phonologically. Here I explore the idea that word order
9 is, at least in part, phonologically determined as well. In section 2.3, I
10 proposed that the position of adverbs is due to phonological linearization.
11 Here I explore the idea that phonology is also responsible for other word
12 order effects. The idea that displacements can be ‘stylistic’ or phonologi-
13 cal originates with Ross, 1967, who initiated the discussion of scrambling
14 within generative grammar and viewed it as a stylistic rule of permuta-
15 tion. More recent accounts of phonological displacement can be found
16 in Adger 1997, Chomsky, 2001b, Kidwai, 1999, McCloskey, 1999, and
17 Zubizarreta, 1998, among others. Prime candidates for displacement in
18 the phonology are heavy NP shift and other stylistic rules which are sen-
19 sitive to phonological properties, e.g., constituent heaviness, adjacency,
20 and movement to an ‘edge’ on the one hand, and movement motivated
21 by focus or topic properties of the moved constituent on the other.

22 A common argument against phonological displacement is its sen-
23 sitivity to island constraints. The violation of island constraints has tradi-
24 tionally been used (e.g., Chomsky, 1977) as a test for wh-movement, that
25 is, purely syntactic A'-movement. However, island constraints, I have
26 argued (1973, 97), are best accounted for as processing constraints sensi-
27 tive to focus structure. If island constraints are formulated as constraints
28 on PF, then rather than arguing against phonological dislocation, they
29 provide evidence for it.⁴⁸ The constraint proposed in Erteschik-Shir,
30 1997 constrains not only wh-trace dependencies, but also anaphoric and
31 negative dependencies among others. It states that such dependencies are
32 restricted to the unmarked f-structure in (70) in which X is a dependent.

34
35 ⁴⁸ The fact that resumptive pronouns ‘save’ islands has been employed to argue that island
36 constraints are, in fact, a PF phenomenon (see Bošković 2001, 114–5 and the references
37 cited therein).

- 1 (70) SUBJECT_{top} [... X ...]_{foc}
 2 sTOP_t

3 According to this constraint, dependencies are restricted to f-structures
 4 in which either the subject or a stage is the topic and the dependent is
 5 contained in the focus.⁴⁹ The intuition behind this constraint is that the
 6 burden of processing a dependency is eased if f-structure and syntactic
 7 structure are aligned. Such alignment is disrupted when the object is the
 8 topic since the remainder of the sentence does not form a syntactic
 9 constituent. This processing constraint can be viewed as part and parcel
 10 of ID since it identifies the syntactic role of the dislocated constituent.
 11 ID should clearly be extended to include complex sentences along these
 12 lines.

15 6.2. *Arguments for phonological displacement*

17 My explanation for the linkage between OS to V-2 no longer hinges on a
 18 phonological account of V-2. It follows rather from an account of the
 19 phonological linearization of adverbs as part of spell-out: If an adverb
 20 intervenes at a required PI site, the string will not be pronounceable. In
 21 view of the fact that there is no reason for V-2 to be ordered with respect
 22 to either adverb placement or PI (as shown in section 2.1), the current
 23 proposal is compatible with both a syntactic and a phonological account
 24 of V-2. Yet there are other well-known reasons to believe that V-2 (and
 25 head movement in general) is not syntactic. Chomsky, 2001b offers sev-
 26 eral. The first reason is that head movement has no semantic effects.
 27 (Within the generally-assumed architecture only syntactic dislocation
 28 can affect interpretation since it provides the input to LF; in an architec-
 29 ture without LF, this argument is not decisive.) Second, overt head-
 30 raising is phonologically motivated by the phonetically affixal character
 31 of inflectional categories. Third, syntactic head-raising is a countercyclic
 32 adjunction rule with different locality constraints. Finally, there is no nat-
 33 ural syntactic notion “second.” Even though only the last reason is theory

36 ⁴⁹ A stage topic specifies the here-and-now of the discourse. It is contextually indexed for
 37 time and place. See Erteschik-Shir, 1997 for details.

1 independent, I would like to pursue what a phonological account of V-2
2 would look like.

3 According to the principles of phonological displacement outlined in
4 section 2.1, the target must be an edge. In order for the verb to be placed
5 in second position, topicalization must follow the fronting of the verb. A
6 phonological account of V-2 therefore requires that topicalization also be
7 phonological. There are independent reasons to believe that topicalization
8 is not syntactic. Topicalization is genuinely optional, since topics can be
9 interpreted in situ. In Danish, for example, there is a preference for top-
10 icalization, but only if there is no violation of ID. It follows that pro-
11 nouns are more often topicalized than are full DPs. Topicalization is
12 also restricted by ID, a PF processing constraint. And topicalization is
13 motivated by processing ease: fronting facilitates the identification of the
14 topic, as noted in section 4, by providing an initial link to previous dis-
15 course. The fact that topicalization is sensitive to island constraints now
16 follows from the extended ID constraint. It is therefore possible to imple-
17 ment the phonological fronting of topics.

18 What, then, is the motivation for phonological V-2? The motivation
19 must be morpho-phonological. Within minimalism the motivation for V
20 to I movement is the affixal character of inflectional categories. Yet this
21 approach does not distinguish overt affixes from covert inflectional fea-
22 tures and is therefore not genuinely phonological. Under the current
23 view, motivation for phonological processes is limited to ease of pronun-
24 ciation and ease of processing where ease of processing (ID) is condi-
25 tioned by the availability of overt morphology. At this point, I can only
26 speculate as to what motivates the movement of the verb to the left edge
27 of its string: In all cases (except for *yes-no* questions), the verb ends up in
28 second position, since a topic, a *wh*-element, or some other element
29 moves in front of it. Abstracting away from questions, the verbal element
30 therefore demarcates the border between the topic and the semantic pred-
31 icate with the resulting form mimicking subject-VP structure in main
32 clauses. This is quite similar to Holmberg, 1986 who proposes that the
33 function of V-2 is to ensure that *C'/CP* will be able to function as a pred-
34 icate, that is, a constituent with a verbal head. Topicalization and V-2
35 thus combine to form an *f*-structure which is isomorphic to syntactic
36 structure, the optimal solution to cases in which the subject-predicate
37 structure does not match the topic-focus structure. Topicalization may

1 select an element from a subordinate clause, but the target to which it
 2 moves is the left edge of the main clause (or main assertion in the case of
 3 bridge verbs). V-2 is therefore also restricted to main clauses.

4 An interesting spin-off of this line of reasoning is that it can potentially
 5 explain why topicalization is not generally employed in English. Since
 6 English does not allow V-2, topicalization results in a non-optimal string.

7 Although my explanation for OS is compatible with syntactic topicali-
 8 zation and verb movement, I have provided some reasons that argue that
 9 these processes should be viewed as phonological. This line of thinking
 10 raises the question of how to distinguish genuinely syntactic dislocation
 11 from phonological dislocation. This is the topic of the next section.

12 13 14 6.3. *Is there any syntactic displacement?*

15
16 In the previous section, I argued that V-2 is a phonological process
 17 mainly in view of its interaction with topicalization. *Yes-no* questions
 18 also involve V-movement. Since it is uneconomical to assume two sepa-
 19 rate types of verb dislocation, one in the syntax and one in the phonol-
 20 ogy, I conclude that all cases of V-movement are phonological.⁵⁰ But if
 21 V-2 in questions is phonological it follows that wh-movement must be so
 22 too, since a phonological implementation of V-2 must target the left edge
 23 and any movement to 'first' position must follow it. This should not be a
 24 surprising result, since topicalization and wh-movement have both been
 25 subsumed under A'-movement, i.e., they have the same properties. Since
 26 their adherence to island constraints is no longer a problem, there is no
 27 reason to force a syntactic analysis. Wh-movement in questions is moti-
 28 vated by the need for the shape of questions to differ (significantly) from
 29 that of assertions. And this difference should come as early as possible in
 30 the string in order to facilitate the processing of the utterance as a ques-
 31 tion. The argument does not carry over to embedded wh-questions and
 32 relative clauses.

33
34
35 ⁵⁰ In fact, I assume no other cases of V-movement: V-I is not needed, since handling inflec-
 36 tion morphologically within PF is consistent with the current view. (See also Williams,
 37 2000 for this view.) V-raising in Larsonian shells is argued against in Erteschik-Shir and
 Rapoport to appear; in preparation.

1 The discussion so far has been limited to main clauses. In order to ac-
 2 count for the phonology of embedding I assume some version of phases à
 3 la Chomsky, 1999. Let us say that a phase is defined as the maximal ex-
 4 tended V projection. This would allow for a flexible view of phases which
 5 can be either VP, IP or CP, depending on what has been merged. Since
 6 each such phase comes with its own edges, these will be the targets of
 7 wh-movement in embedded clauses. But what is the motivation for em-
 8 bedded wh-movement? First, consider again main clause questions. I
 9 have suggested that they must be phonologically signaled. I propose that
 10 a Q feature may be selected in the numeration, which can adjoin only to
 11 main clauses. The Q feature enables a question interpretation and triggers
 12 V-movement. It also functions to block topicalization of elements other
 13 than wh-constituents and it ‘occupies’ first position in *yes-no* questions
 14 leaving the verb in second position across sentence types.

15 Direct questions are thus signaled by inversion triggered by Q and not
 16 the fronting of wh-elements. Moreover, leaving the wh-element in situ
 17 would simplify processing in that there would be no need for ID, the ex-
 18 istence of a variable in the string must also be marked. One way to do
 19 that would be by stress. But wh-elements are not themselves focused (ex-
 20 cept in echo questions, in which they are stressed and remain in situ), so
 21 this is not an option.⁵¹ Motivation for the movement of wh-elements (in
 22 main and embedded clauses) may simply be that this is the only way to
 23 highlight their existence in the string. In cases of multiple wh-phrases,
 24 some languages front only the first one. In Erteschik-Shir, 1997 I show
 25 that the fronted wh-phrase functions as a topic whereas the others are
 26 focused and are therefore independently highlighted. It follows that they
 27 need not front. Other languages require the fronting of all wh-phrases, or
 28 leave them all in situ. It would be interesting to explore whether the wh-
 29 parameter results from different phonological ‘high-lighting’ mechanisms
 30 across languages.

31 I have given a very rough outline of what a phonological account of
 32 V-movement, topicalization and wh-movement would look like. I have
 33 also suggested types of motivation for phonological dislocation. The
 34 dislocation types I have examined do not include A-movement. In
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 37 ⁵¹ See Erteschik-Shir, 1986; 1997 for an analysis of the f-structure of questions.

Erteschik-Shir and Rapoport, in preparation, we argue that passives are directly merged and do not involve movement. Other cases of argument raising may also turn out to have plausible phonological analyses:

- (71) a. John/??a stranger seems to be in the garden.
 b. There seems to be a stranger in the garden.
 (72) a. John/??a stranger seems to have fallen in love with Mary.
 b. It seems that a stranger has fallen in love with Mary.

(71)a and (72)a show that a raised indefinite is degraded. This points to the topic status of raised subjects.⁵² (71)b and (72)b show that in each case an alternate unraised structure is available. Since f-structure linked optionality is characteristic of phonologically motivated dislocation, under the current proposal, it might turn out that DP displacement can also be phonologically implemented.

In this section, I have taken the phonological view of syntax to its extreme, leaving little, if anything for syntax proper. Since this exercise raises interesting questions, it may be worthwhile to pursue this line of investigation further.

7. Conclusion

Although Holmberg, 1999 casts his account of OS in terms of adjacency, he rejects a purely morpho-phonological account since it does not make reference to phonological or morphological primitives. My aim in this paper was to take Holmberg's challenge seriously and to pursue an account which exclusively makes reference to such primitives. My argument is not for or against a particular account of OS, but rather for the plausibility of accounting for word order by purely phonological means. The morpho-phonological primitives I suggest include phonological features such as prosodic properties as well as processing features such as overt case. I believe that the strength of such an approach lies in its potential to explain cross-linguistic and dialectal variation on the basis of fine

⁵² See Erteschik-Shir, 1997 for why modifying an indefinite makes it a better topic as in (i):

i. A gorgeous man seems to have captured Mary's attention.

1 morpho-phonological distinctions between languages, which exist any-
2 way, while maintaining a uniform syntax.

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